✅ Step 1: Dataset Selection (Marketing + Customer Behavior)

A perfect dataset for this is:

🎯 **Online** Retail **II Dataset (UCI / Kaggle)**

- Real e-commerce transactional data
- Contains:
  - Invoice number
  - Customer ID
  - Country
  - Product info
  - Quantity, Price
  - Date of purchase

📥 Download here:
👉 Online Retail Dataset (Kaggle)

Why it's great:

- Ideal for **customer segmentation**
- Can calculate **RFM (Recency, Frequency, Monetary)**
- Fits SQL & Power BI workflows perfectly

✅ Step 2: Roadmap to Build the Project

◆ Phase 1: Define Business Questions

These are the kinds of questions your dashboard will answer:

- Who are my **top customers**?
- What is the **repeat purchase rate**?

Repeat Purchase Rate (%) =

VAR TotalCustomers =

  DISTINCTCOUNT('public customer_rfm'[customerid])

```
VAR RepeatCustomers =

    CALCULATE(

        DISTINCTCOUNT('public customer_rfm'[customerid]),

        FILTER(

            VALUES('public customer_rfm'[customerid]),

            CALCULATE(MAX('public customer_rfm'[frequency])) > 1

        )

    )

RETURN

    DIVIDE(RepeatCustomers, TotalCustomers, 0)
```

- Which **products or categories** are most popular?
- How much revenue comes from **new vs repeat** customers?
- What is the **trend** in customer acquisition and retention?

  Customer Retention
- 🎯 Goal:
  Measure how many customers came back and made purchases in later periods.
- 🔄 Option A: Simple Repeat Buyer Flag (Monthly)
- Create a retention flag per month:
- dax
- CopyEdit
- Repeat Buyer =
- VAR FirstPurchase =
-     CALCULATE(MIN('Sales'[Date of Purchase]), ALLEXCEPT('Sales', 'Sales'[Customer ID]))
- RETURN
-     IF('Sales'[Date of Purchase] > FirstPurchase, 1, 0)
- This flags all purchases after the first as repeat purchases.

- 📊 Retention Rate per Month (Optional Measure)
- You can then create:

- dax
- CopyEdit
- Retention Rate (%) =
- VAR TotalCustomers = CALCULATE(DISTINCTCOUNT('Sales'[Customer ID]))
- VAR RepeatCustomers =
-     CALCULATE(
-         DISTINCTCOUNT('Sales'[Customer ID]),
-         FILTER('Sales', [Repeat Buyer] = 1)
-     )
- RETURN
-     DIVIDE(RepeatCustomers, TotalCustomers, 0)

- What is the **average order value** and how is it changing?

◆ Phase 2: Data Prep (SQL Work)

Use SQL to:

- Clean and filter data (remove canceled orders, nulls, etc.)
- Create tables/views for:
    - Customer purchase frequency
    - Recency (last purchase date)
    - Total revenue per customer (Monetary)
    - Monthly sales trend
    - Product category-level performance
- Assign **RFM scores** to segment customers (RFM = Recency, Frequency, Monetary)

Save each major SQL query or transformation — you'll include this in the case study later.

◆ Phase 3: Power BI Dashboard (Frontend)

Create an interactive dashboard with:

1. **KPI cards**: Total Revenue, Avg Order Value, Repeat Rate
2. **RFM Customer Segmentation** (Donut or Cluster Chart)
3. **Revenue Trend** (Line graph: monthly/yearly)
4. **Sales by Country or Product Category** (Map or bar chart)
5. **Top 10 Customers** (Bar chart with customer names and total spend)
6. **Filters**: Country, Date, Product, Customer Type

💡 **Use slicers** and interactive drilldowns to make the dashboard feel dynamic and "client-ready."

✅ Step 3: Case Study Write-Up Format

Here's what to include:

📄 Customer Behavior & Marketing Insights Dashboard

Portfolio Case Study Example

---

1. Objective

To analyze customer purchasing behavior and generate insights to improve marketing campaigns, customer retention, and product targeting.

---

2. Dataset Overview

Online Retail dataset with transactional data from 2010–2011, including Customer ID, Purchase Date, Country, Product, Quantity, and Price.

---

3. Approach

- Used SQL to clean and transform raw sales data.

- Calculated RFM metrics (Recency, Frequency, Monetary) to segment customers.
- Built interactive Power BI dashboards to display revenue trends, top customers, repeat purchase behavior, and regional sales insights.
- Enabled filters by product, date, customer type, and country.

---

4. Key Insights

- 20% of customers generated 60% of total revenue.
- Repeat customers had a 45% higher average order value than new customers.
- UK and Germany were top markets, with seasonal sales spikes in November–December.

---

5. Outcome

Marketing and product teams can use this dashboard to:

- Identify high-value customer segments
- Tailor loyalty campaigns to top spenders
- Optimize promotions for high-performing regions and categories

---

6. Tools Used

SQL (data cleaning and transformations)
Power BI (data modeling, visual storytelling)
DAX (for calculated measures)

---

7. Optional

- Include a **Power BI screenshot** or a **link to the published report**
- Add a **PDF download** version of the case study

Would you like help with:

- Example **SQL queries** for this dataset (RFM, customer analysis)?
- Mockup layout of your Power BI dashboard visuals?

Let me know how you'd like to continue — this project is a great one to showcase both your analytical thinking and tool expertise!

4o

You said:

can you help me with the SQL work for the above dataset?

ChatGPT said:

Absolutely! Here's a step-by-step **SQL guide** to prepare and analyze the **Online Retail dataset** for your Power BI project on **Customer Behavior and Marketing**.

Assumption: You're working with a table called online_retail, with the following relevant fields:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

✅ STEP 1: Data Cleaning

◆ Remove returns and nulls

sql
CopyEdit
```
SELECT *
FROM online_retail
WHERE Quantity > 0
```

```
  AND UnitPrice > 0
  AND CustomerID IS NOT NULL;
```

✅ STEP 2: Create a Cleaned Sales Table (CTE or View)

sql
CopyEdit
```
CREATE VIEW clean_sales AS
SELECT
  InvoiceNo,
  CustomerID,
  Country,
  InvoiceDate,
  Description,
  Quantity,
  UnitPrice,
  Quantity * UnitPrice AS TotalAmount
FROM online_retail
WHERE Quantity > 0
  AND UnitPrice > 0
  AND CustomerID IS NOT NULL;
```

✅ STEP 3: RFM Calculation (Recency, Frequency, Monetary)

◆ 1. Define latest date for reference (e.g., max invoice date)

sql
CopyEdit
```
SELECT MAX(InvoiceDate) AS LatestDate FROM clean_sales;
```

Let’s say the LatestDate is '2011-12-09'.

◆ 2. RFM Table

sql
CopyEdit
```
SELECT
```

  CustomerID,
  DATEDIFF(DAY, MAX(InvoiceDate), '2011-12-09') AS Recency,
  COUNT(DISTINCT InvoiceNo) AS Frequency,
  SUM(TotalAmount) AS Monetary
FROM clean_sales
GROUP BY CustomerID;

You can use this to segment customers in Power BI using **buckets or quartiles** for R, F, M.

✅ **STEP 4: Monthly Sales Trend**

sql
CopyEdit
```
SELECT
  FORMAT(InvoiceDate, 'yyyy-MM') AS SalesMonth,
  SUM(TotalAmount) AS MonthlyRevenue
FROM clean_sales
GROUP BY FORMAT(InvoiceDate, 'yyyy-MM')
ORDER BY SalesMonth;
```

✅ **STEP 5: Country-Level Sales Performance**

sql
CopyEdit
```
SELECT
  Country,
  SUM(TotalAmount) AS TotalRevenue,
  COUNT(DISTINCT CustomerID) AS UniqueCustomers
FROM clean_sales
GROUP BY Country
ORDER BY TotalRevenue DESC;
```

✅ **STEP 6: Top 10 Customers by Revenue**

sql
CopyEdit

```
SELECT TOP 10
  CustomerID,
  SUM(TotalAmount) AS TotalSpent,
  COUNT(DISTINCT InvoiceNo) AS Orders
FROM clean_sales
GROUP BY CustomerID
ORDER BY TotalSpent DESC;
```

✅ STEP 7: First-Time vs Repeat Customers

Create a CTE to find the first purchase date:

sql
CopyEdit

```
WITH FirstPurchase AS (
  SELECT
    CustomerID,
    MIN(InvoiceDate) AS FirstPurchaseDate
  FROM clean_sales
  GROUP BY CustomerID
)
SELECT
  c.CustomerID,
  CASE
    WHEN DATEDIFF(DAY, fp.FirstPurchaseDate, c.InvoiceDate) = 0 THEN 'New'
    ELSE 'Repeat'
  END AS CustomerType,
  SUM(c.TotalAmount) AS Revenue
FROM clean_sales c
JOIN FirstPurchase fp ON c.CustomerID = fp.CustomerID
GROUP BY c.CustomerID, fp.FirstPurchaseDate, c.InvoiceDate;
```

✅ STEP 8: Create Power BI Views

Save the important queries as SQL **views or output tables** so that Power BI can connect directly:

- rfm_scores
- monthly_sales

- customer_segmentation
- country_revenue

🔄 **Next Steps**

Now you can import these views/tables into Power BI and:

- Build a data model
- Create DAX measures for KPIs
- Build the dashboard visuals

Would you like help writing the DAX measures or setting up the Power BI visuals next?

4o

You said:

yes

ChatGPT said:

Perfect! Now that your SQL prep is done, let's move into **Power BI** to build the visuals and DAX measures for your **Customer Behavior & Marketing Dashboard**.

✅ **Step 1: Import SQL Views or Queries**

In Power BI Desktop:

1. Click **Get Data** → **SQL Server** (or the DB you used)
2. Connect to:
   - clean_sales
   - rfm_scores
   - monthly_sales
   - customer_segmentation
   - country_revenue

✅ Step 2: Create Relationships (if needed)

- CustomerID is your key column — link all tables/views using it.
- Link monthly_sales to a calendar table (if you have one).
- Create a **Date table** using DAX if needed:

DAX
CopyEdit
```
DateTable = CALENDAR(MIN(clean_sales[InvoiceDate]),
MAX(clean_sales[InvoiceDate]))
```

✅ Step 3: Create DAX Measures

Here are the key DAX measures you'll need:

◆ 1. Total Revenue

DAX
CopyEdit
```
Total Revenue = SUM(clean_sales[TotalAmount])
```

◆ 2. Average Order Value

DAX
CopyEdit
```
Average Order Value =
DIVIDE(
   SUM(clean_sales[TotalAmount]),
   DISTINCTCOUNT(clean_sales[InvoiceNo])
)
```

◆ 3. Total Orders

DAX
CopyEdit

Total Orders = DISTINCTCOUNT(clean_sales[InvoiceNo])

◆ 4. Unique Customers

DAX
CopyEdit
```
Unique Customers = DISTINCTCOUNT(clean_sales[CustomerID])
```

◆ 5. New vs Repeat Revenue

Assuming you have a column CustomerType in a customer_segmentation table:

DAX
CopyEdit
```
New Customer Revenue =
CALCULATE(
    SUM(customer_segmentation[Revenue]),
    customer_segmentation[CustomerType] = "New"
)

Repeat Customer Revenue =
CALCULATE(
    SUM(customer_segmentation[Revenue]),
    customer_segmentation[CustomerType] = "Repeat"
)
```

◆ 6. RFM Labels (optional)

If your rfm_scores has Recency, Frequency, and Monetary scores, you can classify:

DAX
CopyEdit
```
RFM Segment =
SWITCH(
    TRUE(),
```

```
  rfm_scores[Recency] <= 30 && rfm_scores[Monetary] >= 1000, "Best
Customers",
  rfm_scores[Recency] > 90 && rfm_scores[Monetary] < 500, "At Risk",
  rfm_scores[Recency] <= 60 && rfm_scores[Frequency] >= 5, "Loyal
Customers",
  "Other"
)
```

✅ Step 4: Suggested Dashboard Layout

📊 Page 1: Executive Overview

- **KPI Cards:** Total Revenue, Total Orders, AOV, Repeat Revenue %
- **Donut Chart:** New vs Repeat Customer Revenue
- **Bar Chart:** Top 10 Customers
- **Line Chart:** Monthly Revenue Trend

📊 Page 2: Marketing Insights

- **Clustered Bar:** Revenue by Country
- **Treemap:** Revenue by Product Category (if you added this in SQL)
- **Heatmap/Table:** RFM Segments with # of Customers and Revenue

---

📊 Page 3: Customer Segmentation

- **Table:** RFM scores with Customer IDs
- **Slicers:** Country, Date Range, Customer Type

✅ Step 5: Bonus — Publish and Share

Once complete:

- Save and **publish** to Power BI Service (you can use “Share to Web”)
- Export your visuals to PDF
- Add screenshots + summary to your **portfolio case study**

Would you like me to:

- Help you design a **case study PDF template**, or
- Review your DAX and visual design for polish?

1. Clean and Transform (Optional)

- Click **“Transform Data”** to open Power Query Editor
- Clean column names, filter rows, change data types if needed

2. Build a Simple Dashboard

Start with common visuals like:

- **Card**: Total revenue or total orders
- **Bar chart**: Top 10 customers by spend
- **Pie chart**: Sales by country or category
- **Line chart**: Sales trend over time
- **Slicer**: Filter by month, product, customer, etc.

3. Create Measures (Optional)

Use DAX formulas to create custom metrics, for example:

DAX
CopyEdit
```
Total Revenue = SUM(Sales[TotalAmount])
Average Order Value = [Total Revenue] / COUNT(Sales[OrderID])
```

4. Format & Publish

- Add titles, tooltips, and legends
- Format currency, dates, labels
- Save the report
- Optional: Publish to Power BI Service if you want to share

Set Your Dashboard Goal

Let's focus on key business questions:

- Who are our best customers?
- What do they buy and when?
- What's the sales trend?
- Which products are performing well?
- Where are the customers from?

---

2. Suggested Visuals & Layout

Here's a sample layout with relevant visuals:

◆ **Top Section – KPI Cards**

Use **Card visuals** for high-level metrics:

- **Total Revenue**
- **Total Customers**
- **Total Orders**
- **Average Order Value**

Use icons and color-coded backgrounds for visual interest.

◆ **Left Column – Customer Insights**

1. **Top 10 Customers by Revenue**
   - Visual: **Horizontal bar chart**
   - Sort by: Descending revenue
2. **Customer Segment by Spend**
   - Visual: **Donut or Treemap**
   - Compare segments (new vs returning, region, etc.)

◆ **Right Column – Sales Insights**

1. **Sales Over Time**
   - Visual: **Line chart**

   - Use Order Date vs Total Sales
2. **Orders by Category/Product**
   - Visual: **Stacked bar chart or Clustered column**

◆ **Bottom – Filters and Slicers**

- 📅 **Date range slicer**
- 🛒 **Product Category slicer**
- 🌍 **Country slicer**

---

3. Tips for Visual Appeal

🎨 **Design**

- Use a clean, consistent color palette (e.g., blue/orange/grey)
- Align visuals into a **grid layout**
- Limit visuals per page (5–7 max) for clarity
- Add **icons** (use Insert → Shapes or Images)
- Use **“Title” and “Tooltips”** for every visual

Usability

- Use **hierarchies** (e.g., Year > Month) for drill-down
- Set **“Sorting”** properly (e.g., by value or date)
- Use **conditional formatting** to highlight top/bottom values

📐 **Best Practices**

- Keep filters visible (slicers or filter pane)
- Add a **report title and update timestamp**
- Use **DAX measures** instead of calculated columns where possible

---

📌 **Example DAX Measures**

DAX
CopyEdit

Total Revenue = SUM(Sales[TotalAmount])
Total Orders = COUNT(Sales[OrderID])
Average Order Value = [Total Revenue] / [Total Orders]

Recommended DAX Measures

Assume your table is called Sales with columns like:

- OrderID
- CustomerID
- TotalAmount
- OrderDate
- Country
- ProductCategory

DAX
CopyEdit

```
Total Revenue = SUM(Sales[TotalAmount])

Total Orders = DISTINCTCOUNT(Sales[OrderID])

Total Customers = DISTINCTCOUNT(Sales[CustomerID])

Average Order Value = [Total Revenue] / [Total Orders]

Monthly Revenue =
CALCULATE(
   [Total Revenue],
   DATESINPERIOD(
      Sales[OrderDate],
      MAX(Sales[OrderDate]),
      -12,
      MONTH
   )
)
```

You can also add:

DAX
CopyEdit

```
Top Customer =
TOPN(
    1,
    SUMMARIZE(Sales, Sales[CustomerID], "Revenue", [Total Revenue]),
    [Total Revenue],
    DESC
)
```

🏷️ Suggested Titles for Visuals

Visual	Title
Card 1	💰 Total Revenue
Card 2	📦 Total Orders
Card 3	👥 Total Customers
Card 4	Avg Order Value
Bar Chart	🏆 Top 10 Customers by Revenue
Line Chart	📈 Monthly Sales Trend
Stacked Column	🔍 Orders by Product Category
Map or Bar (Country)	🌍 Revenue by Country
Slicers	🎚️ Filter by Date, Category, etc.

🚀 1. Customer Journey Funnel

Visual Type: Funnel Chart
What to Show: Stages from site visit → add to cart → purchase → repeat customer
How:

- Use conversion step columns or flags (can be simulated with calculated columns if exact data is not present)
- Use DAX to count unique customers in each stage

Visual Impact: Tells the drop-off points in the customer journey.

📍 2. Geo Heat Map of Sales by Country/City

Visual Type: Filled Map or Shape Map
What to Show: Revenue or Order volume by country/region
Enhancement: Add a slicer to filter by product category or customer segment

Visual Impact: Geographic trends in customer behavior

📊 3. Customer Segmentation Matrix

Visual Type: Matrix or Heatmap
What to Show:

- Rows: Customer Segment (e.g., new, repeat, VIP, churn-risk)
- Columns: Metrics like Avg Order Value, Lifetime Value, Order Frequency

Bonus: Color-code with conditional formatting
How: Use DAX to create customer tags/segments

🔁 4. Retention or Repeat Purchase Analysis

Visual Type: Line or Bar Chart
What to Show: % of customers who made 2+ purchases in a time period
How: Use CustomerID with OrderDate to calculate first purchase vs repeat purchases

Visual Impact: Shows stickiness and customer loyalty

💬 5. Dynamic Tooltips with Mini KPIs

Visual Type: Tooltip Pages
What to Show: Hover over a region/customer to show:

- Total Orders
- Avg Order Value
- Products Purchased
- Last Purchase Date

How: Create a tooltip report page in Power BI and assign it to a visual

---

6. AI Insights (Decomposition Tree or Key Influencers)

Visual Type: Key Influencers / Decomposition Tree
What to Show:

- What drives high revenue?
- What influences repeat purchases?

Visual Impact: Shows data discovery and intelligence

---

7. Smart Interactions & Slicers

Add:

- Toggle between Revenue and Orders (via bookmarks or field parameters)
- Slicers for Product Category, Customer Segment, Region
- Time period toggles (YTD, Last 30 Days, etc.)

---

8. Experiment: Predictive Trends or Forecasting

Use Power BI's built-in **forecasting** on a line chart:

- Project future sales by month
- Add confidence intervals

Bonus Titles and Layout Tips

Dashboard Title:
Customer Behavior and Sales Insights Dashboard
Understand who buys, how often, and what drives repeat revenue

Sections:

- KPIs at the top (Revenue, Orders, Repeat Rate)
- Segments in the middle (customer types, products)
- Time & geography at the bottom
- Always leave room for storytelling annotations

📈 1. Customer Acquisition

🎯 Goal:

Count **new customers** acquired over time (e.g., per month or year).

✅ Step-by-step:

Create a measure or calculated column for First Purchase Date:

dax
CopyEdit
```
First Purchase Date =
CALCULATE(
   MIN('Sales'[Date of Purchase]),
   ALLEXCEPT('Sales', 'Sales'[Customer ID])
)
```

This gives the **first purchase date per customer**.

Then create a new column to extract the acquisition month or year:

dax
CopyEdit
```
Customer Acquisition Month =
FORMAT([First Purchase Date], "YYYY-MM")
```

You can now use a **bar/line chart**:

- Axis: Customer Acquisition Month

- Values: DISTINCTCOUNT(Customer ID)

This shows how many **new customers** you acquired each month.

🔁 2. Customer Retention

🎯 Goal:

Measure how many customers came back and made purchases in later periods.

🔄 Option A: Simple Repeat Buyer Flag (Monthly)

Create a **retention flag** per month:

dax
CopyEdit

```
Repeat Buyer =
VAR FirstPurchase =
   CALCULATE(MIN('Sales'[Date of Purchase]), ALLEXCEPT('Sales',
'Sales'[Customer ID]))
RETURN
   IF('Sales'[Date of Purchase] > FirstPurchase, 1, 0)
```

This flags all purchases **after the first** as repeat purchases.

📊 Retention Rate per Month (Optional Measure)

You can then create:

dax
CopyEdit

```
Retention Rate (%) =
VAR TotalCustomers = CALCULATE(DISTINCTCOUNT('Sales'[Customer ID]))
VAR RepeatCustomers =
   CALCULATE(
      DISTINCTCOUNT('Sales'[Customer ID]),
      FILTER('Sales', [Repeat Buyer] = 1)
```

```
    )
RETURN
    DIVIDE(RepeatCustomers, TotalCustomers, 0)
```

Bonus: Cohort Analysis

Use Customer Acquisition Month and compare behavior over following months. Let me know if you'd like to build that — it's very powerful but requires a few extra steps.

Summary

Metric	Description	How to Visualize
First Purchase Date	When each customer first bought	Line/Bar by month
Acquisition Count	# of new customers each month	Line/Bar chart
Repeat Buyer Flag	Whether a purchase is repeat or not	Filtered table or slicer
Retention Rate	% of customers making repeat purchases	KPI, card, or line over time

---

Would you like help setting up a **cohort retention matrix** (e.g., Month 0, Month 1, Month 2...) to impress even more in your dashboard?